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How do SMEs' strategies change under the impacts of a critical external shock? The case of Vietnamese SMEs under the COVID-19 pandemic

Long Hoang Le, An Hoang Kim Vo, Duong Tuan Nguyen, Giang Huong Duong

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How do SMEs' strategies change under the impacts of a critical external shock? The case of Vietnamese SMEs under the COVID-19 pandemic

Long Hoang Le

School of International Business and Marketing,
University of Economics Ho Chi Minh City,
Ho Chi Minh City, Vietnam
Email: longlh@ueh.edu.vn

An Hoang Kim Vo*

College of Management,
National Sun Yat-sen University, Taiwan
and
Foreign Trade University,
Ho Chi Minh City Campus, Vietnam
Email: vohoangkiman.cs2@ftu.edu.vn
*Corresponding author

Duong Tuan Nguyen

College of Management,
National Sun Yat-sen University, Taiwan
and
Foreign Trade University,
Ho Chi Minh City Campus, Vietnam
Email: nguyentuanduong.cs2@ftu.edu.vn

Giang Huong Duong

Faculty of Business Administration,
Ho Chi Minh City Open University,
Ho Chi Minh City, Vietnam
Email: giang.dh@ou.edu.vn

Abstract: The impacts of external shock caused by the coronavirus (COVID-19) pandemic force small and medium enterprises (SMEs) to change their strategy to survive and/or even grow. This study aims to explore how SMEs change their strategies under the impact of the pandemic. We used semi-structured in-depth interviews with owners/managers of SMEs in Vietnam immediately after the outbreak of the pandemic and the implementation of the social distancing policy to collect the data. A thematic analysis was carried out under the theoretical lens of the dynamic capability's theory and strategic responses to the crisis model. The findings revealed the crucial role of dynamic capabilities with SMEs' strategy changes during the acute social crisis. In

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particular, firms with innovative capability and agility capability employed innovative strategies to exploit opportunities from the uncertain market. Meanwhile, firms with less preparation for crisis management adopted a perseverance strategy, and the firms without those characteristics followed retrenchment strategies.

Keywords: COVID-19; dynamic capabilities theory; pandemic; SMEs; Vietnam.

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Biographical notes: Long Hoang Le holds a PhD degree at RMIT Vietnam. He works as a Lecturer at University of Economics Ho Chi Minh City. His research interests are consumer behaviour in online community, technology adoption behaviour, and health information management. He has published in *Journal of Brand Management*, *Computers in Human Behavior*, *Health Communication*, *Journal of International Consumer Marketing*, *Gender and Technology Development*, and several IS prestige conferences such as ACIS, TSWIM, and INFORMS annual meeting.

An Hoang Kim Vo is a young Lecturer at Foreign Trade University, Vietnam. She accomplished a Master's degree at La Trobe University, Melbourne, Australia. Then, she was offered a scholarship by the National Sun Yat-Sen University, Kaohsiung, Taiwan, for the PhD student position. Her fascinating research field is organisational behaviour, especially decision making, emotions, and personalities in teams and organisations.

Duong Tuan Nguyen is a young Lecturer at Foreign Trade University, Vietnam. He got the scholarship from Taiwan Ministry of Education (MOE) for his MBA majoring in International Business at National Sun Yat-sen University, Taiwan. From 2018, he is pursuing his PhD program at the College of Management, National Sun Yat-sen University, Taiwan. His research interests contain topics relating to business administration especially business strategy and organisational behaviours.

Giang Huong Duong is a Lecturer at HCMC Open University, Vietnam. She is a PhD candidate at RMIT University Vietnam. Her research interests are consumer behaviour, security behaviour, and supply chain management. She has published in *Journal of Brand Management* and *Journal of International Consumer Marketing*.

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1 Introduction

The major external shocks might cause different effects on various aspects of the economy and disrupt economic equilibrium (McKibbin and Fernando, 2021; Ratten, 2020). As a severe health crisis, the COVID-19 pandemic and the social distancing policy have forced extensive changes in both consumer behaviour and firms' operations. The impact of the pandemic is more complicated and severe in emerging economies (Omar et al., 2020; Salamzadeh and Dana, 2021). In addition, the pandemic caused further serious problems for small and medium-sized enterprises (SMEs) due to their resource constraints and vulnerability (Bartik et al., 2020; Juergensen et al., 2020).

It is recognised that SMEs were found to play an important role in promoting the economy due to several idiosyncratic characteristics of flexibility, dynamics, and diversity (Andries and Czarnitzki, 2014; Sharma and Tarp, 2018). According to existing literature, any country's economic prosperity is inextricably connected to the development of SMEs. For instance, as Beck et al. (2005) asserts, there is a substantial and favourable association between the size and number of the SMEs and economic development. Ayyagari et al. (2007) also contends that formal SMEs contribute over half of the GDP in high-income nations on average. In emerging economies, SMEs create jobs for available and abundant low-skilled individuals, and respond quickly to urgent market demand (Lee and Newton, 2000). In Vietnam, the official report of the Vietnamese General Statistics Office (2017) indicated that SMEs account for 98.1% of the total number of businesses operating in 2017, creating more than 6 million jobs, contributing nearly 50% of the GDP, over 30% of the state budget revenue, and 45% of investment capital implemented in the whole economy.

However, it is indicated that SMEs are the most affected group due to their vulnerability under economic instabilities (Bartik et al., 2020; Dyduch et al., 2021; Chan and Mustafa, 2020). They have apprehensively struggled to navigate through the COVID-19 pandemic, but they have received meagre attention during this crisis (Salamzadeh and Dana, 2021). Moreover, SMEs rarely have preparations for crisis situations that contain strategies to deal with various potential scenarios, so they will behave differently when faced with crises because of the complexity and diversity (Pereira et al., 2021; Salamzadeh and Dana, 2021). For example, while the majority of current businesses have been required to cut down on costs (e.g., labour costs, rental costs) to survive the COVID-19 pandemic (Bartik et al., 2020), other companies can take advantage of the opportunity emerging during this time to transform business structure and strategy (Akpan et al., 2020). In short, COVID-19 has been a quintessential 'hot crisis' accompanied by a slew of unprecedented troubles that required firms to deliberate and generate proper and effective coping strategies to survive. The current situation has given rise to questions about how the impacts of the COVID-19 pandemic on SMEs and how SMEs changed their business to deal with this major, sudden and inevitable external shock.

Since the beginning of the COVID-19 outbreak, a significant number of studies have been conducted to investigate the impact of this crisis and how businesses behave in such an acute pandemic (del Rio-Chanona et al., 2020; Juergensen et al., 2020; Ratten, 2020). Still, there has been little research done on this issue that investigates the strategy changes of SMEs in response to the major external shock (Dyduch et al., 2021; Etemad, 2020). Some recent studies focused on investigating SMEs immediate responses to the COVID-19 pandemic in developed countries (Bartik et al., 2020; Juergensen et al., 2020;

Kraus et al., 2020), the responses of SMEs to the government policy (OECD, 2020; Omar et al., 2020), or in some specific business operation (marketing, logistics, operation, and human resource management) (Nayal et al., 2021; Sharma et al., 2020). There is a lack of investigation on SMEs' business coping strategies in emerging economies from the strategic perspective of the owners/managers (Fabeil et al., 2020; Hameed et al., 2021; Herbane, 2019); the situation of SMEs in developing countries is especially vulnerable (Sleuwaegen and Goedhuys, 2002) due to the embryonic and undeveloped financial systems and high information asymmetry (Nguyen et al., 2019).

This study aims to explore the impact of a major external shock, namely the COVID-19 pandemic, on the business strategies of SMEs based on the dynamic capabilities theory. Dynamic capabilities refer to firms' ability to promptly grasp the market fluctuations and seize the opportunity delivered by the business environment (Wilden et al., 2013). Accordingly, this theory appears to be appropriate to justify the change in business strategy; specifically, how firms create their competitiveness, generate profits and adapt to the constantly changing environment (Ambrosini and Bowman, 2009; Helfat et al., 2009; Herbane, 2019). This theory can also be applied to understand the firms' decisions when dealing with the difficult situation sparked by the 'hot crisis' COVID-19. In particular, a detailed discussion on the firm's dynamic capabilities as the fundamental foundation to propose effective coping strategies in confronting the COVID-19 pandemic of SMEs in an emerging market will be presented. The qualitative approach with semi-structured in-depth interviews was employed with the participation of 15 owners/managers of SMEs in Ho Chi Minh City, which is the biggest city and was directly and severely affected by the COVID-19 pandemic in Vietnam. The interview was conducted immediately after the COVID-19 pandemic and the social distancing policy had ended to get more insight into the research topic. The expected results of the present study may provide timely scientific evidence and endeavours to extend the literature on how SMEs recognise the firm's dynamic capacities to have response strategies in dealing with the external shocks namely the COVID-19 pandemic.

This study extends previous research on SMEs and strategic decisions in three ways. First, although many prior studies have attempted to examine how firms respond to the severe consequences of the COVID-19 pandemic using several kinds of dynamic capabilities (e.g., Nayal et al., 2021; Ai and Peng, 2021; Do et al., 2021), these studies lack a common theoretical framework, and therefore the findings of these studies cannot be generalised to all available types of dynamic capabilities. In the current study, the dynamic capability theory (Teece et al., 1997; Wang and Ahmed, 2007), combined with newly proposed ones such as network capability (Parida et al., 2009) and market sensing capability (Jusoh and Parnell, 2008) and their relationship with certain strategic decisions are explored. In doing so, the way that firms capitalise on their dynamic capabilities to confront the COVID-19 crisis can be investigated in a more systematic and thorough manner. Adopting the framework is essential for it might help identify the effect of capabilities that may have been obscured in previous research.

Second, although quite a lot of studies examined how firms respond to the COVID-19 pandemic, the majority of the studies adopted survey or systematic review methods (e.g., Bartik et al., 2020; Juergensen et al., 2020; Hameed et al., 2021). Only a few researchers adopted a qualitative approach (Kraus et al., 2020; Nayal et al., 2021) but do not have a theoretical foundation on any theory. This will lead to equivocal research results due to measurement errors and biases due to the nature of survey research. Given that using a qualitative approach is a viable strategy that is workable in studying SMEs and

entrepreneurial issues (Dana and Dana, 2005), the current study utilised the qualitative approach with semi-structured in-depth interviews to straightforwardly grasp the information from the study subjects and generate a holistic picture regarding how Vietnam SMEs dealt with COVID-19.

Finally, the COVID-19 pandemic has shown detrimental consequences on the world economy. SMEs are the most severely affected because they have limited resources. Nonetheless, they also possess flexibility in the operation. Therefore, examining the response of SMEs in a crisis would provide practical implications for SMEs' business administration and management. Furthermore, SMEs account for the most businesses in every country but are also the most vulnerable to external shocks. Hence, understanding the way SMEs operate and respond to a crisis will assist policymakers in better supporting SMEs to overcome the challenges posed by the COVID-19 pandemic and future crises.

2 Literature review

2.1 COVID-19 as a crisis and firms' crisis management

A 'crisis' refers to the circumstances where "potentially stressful events associated with the emergence of health threats (e.g., epidemics, biological and chemical contamination of food), terrorist attacks, natural disasters (e.g., hurricanes, floods), industrial accidents (e.g., nuclear) or even events related with macroeconomic change" (Gaspar et al., 2016). Despite its common usage, the definition of 'crisis' has several variants across different disciplines. For instance, 'social crisis' in the study of Oh et al. (2013) refers to "the severe consequences of the incident, low probability of incident occurrence, informational and situational uncertainty, and decision-making pressure under time constraints". Although the definition has evolved, there appears to be agreement that 'crisis' situations embody a multitude of characteristics, encompassing unexpected shifts in a system (Greiner, 1989), a menace to firms' survival (Witte, 1981), a plethora of stakeholders are implicated (Elliott and McGuinness, 2002), decision-making under time pressure (Pearson and Clair, 1998), low chance of incidence and harsh detrimental impacts (Hills, 1998), informative ambiguity, and unfavourable sentiments in the community (e.g., anxiety, anger, fear) (Seeger et al., 2003).

Emanating since late 2019, the COVID-19 pandemic has been considered a once-in-a-lifetime event that has caused severe impacts, with over 175 million infected cases and 3.7 million deaths globally (WHO, 2021). Regarding virulence and community transmission (Reed et al., 2013), it has been described as one of the most devastating health epidemics in modern history (Freitas et al., 2020). Due to the virus's high rates of contagion, several countries have enacted 'social distancing' health tactics for the first time, which, in addition to affecting aspects of people's daily life, have had enormous economic repercussions across the world (Kraus et al., 2020). Stock markets have plummeted, and analysts have repeatedly predicted grave economic downturns (Baker et al., 2020; McKibbin and Fernando, 2021). In certain areas, governments have enforced stringent stipulations on firms or even shut down small businesses, resulting in demand-supply complications (del Rio-Chanona et al., 2020). The labour market is amongst the hardest-hit areas, with approximately 22 million Americans laid off in only one month, there is a twofold increase in Austria's unemployment figure, and temporary

staff cutbacks had been imposed on a third of the Swiss workforce (Kraus et al., 2020). In short, COVID-19 is a quintessential 'hot crisis' accompanied by a slew of unprecedented troubles that requires firms to deliberate and generate proper and effective coping strategies to survive.

Concerning the way to cope with the crisis, Coombs (2014) introduced and defined *crisis management* as "a set of strategies formulated to lessen crisis and to lessen damage induced by the crisis". According to Smith (2005), crisis management had two main functions:

1 halting hazardous incidences

2 curbing the deleterious effects of hazardous incidences.

In the same vein, Doern (2016) asserted that crisis management carries certain connotations in some types of mitigating and alleviating crisis repercussions. It was a challenging procedure since it necessitates a multiplicity of adjustments to find a proper resolution to existing difficulties (Nayal et al., 2021). Nevertheless, there was no one-size-fits-all approach to crisis management since it is dependent on a variety of dynamic elements in the environment (Buchanan and Denyer, 2013).

The research on organisational crisis management in regard to the aftermath of a crisis covered a wide range of topics, including changes in liaisons with stakeholders (Pfarrer et al., 2008) or the impact of firms' adaptability and knowledge absorption on their continuity (Lampel et al., 2009). Still, crises did not necessarily have solely detrimental ramifications for those involved. The possible favourable implications of crises were also accentuated in the literature. Particularly, extant research has emphasised that if organisations could recognise opportunities and seize the chance to manage the crisis properly, they might be facilitated to achieve a higher level of prosperity (Faulkner, 2001; Ratten, 2020). Such crises enabled firms to adopt new operational paradigms and undergo several beneficial adjustments, and aid in the stimulation of value creation, innovation, and the identification of new business opportunities (Faulkner, 2001).

In general, crisis management has become a key component in firms' effective managerial strategies, making it imperative for an organisation's long-term viability and sustainability in such a dynamic marketplace (Chong, 2004; Nayal et al., 2021). As a result, businesses should revise their policies regularly to ensure that they survive and thrive in unpredictable times, especially in the severe COVID-19 pandemic. In the literature, various researchers have developed frameworks to address the crisis predicament (Nayal et al., 2021; Smith, 2005). We argue that managers should develop coping strategies based on a firm's dynamic capabilities to encounter adversities stemming from the COVID-19 pandemic effectively.

2.2 Dynamic capabilities theory

Dynamic capabilities were defined as "the firm's ability to integrate, build, and reconfigure internal and external competencies to address rapidly changing environments" (Teece et al., 1997). In a capricious business environment characterised by rapid innovation and global competition, firms need to strengthen their business strategy by continuously analysing, deliberating, and deploying resources (both tangible and intangible) in unstable conditions to gain competitive advantage instead of solely depending on internal resources, and this helps them to compete more effectively with

their rivals (Teece et al., 1997). Therefore, this theory can serve as a solid foundation for businesses to create their competitiveness, generate profits and adapt to the constantly changing environment (Ambrosini and Bowman, 2009; Helfat et al., 2009), especially when dealing with the problematic situation sparked by the 'hot crisis' COVID-19.

Extant research has illustrated the positive effect of dynamic capabilities on a firm's performance and its ability to encounter business environment fluctuations. Particularly, dynamic capabilities, according to Teece (2012), may ultimately contribute to firms' competitive edges and enhanced resilience in the context of vacillating environments. Furthermore, a study of firms in various industries conducted by Makkonen et al. (2014) indicated that dynamic capabilities could augment the demand for new products, indicating their beneficial influence on innovation performance. Indeed, Hang (2016) exerted effort to investigate the underlying mechanism of the positive relationship between dynamic capabilities and a firm's innovation performance and found that opportunity recognition indirectly enhances firm performance through opportunity utilisation. Moreover, dynamic capabilities were found to assist firms in being successful within the context of internalisation (Wu et al., 2016).

Additionally, dynamic capabilities are also related to firms' ability to promptly grasp the market fluctuations and seize the opportunity delivered by the business environment (Wilden et al., 2013). A particular resource can be considered to have dynamic capabilities with four characteristics: *valuable*, *scarce*, *inimitable* and *irreplaceable*. Valuable resources enable firms to fulfil immediate market demand and simultaneously serve as a flexible source that helps businesses leap at the chance and contend with economic and environmental pressures. Scarce resources are rare, existing only in small numbers which are difficult for competitors to possess and are therefore valuable. Meanwhile, inimitable resources are resistant to duplications, and it is impossible to replace.

According to Wang and Ahmed (2007), dynamic capabilities have three fundamental components, including *innovative capability*, *adaptive capability* and *absorptive capability*. Over time, more recent researchers complemented additional abilities, namely network capability (Parida et al., 2009) and market sensing capabilities (Jusoh and Parnell, 2008; Morgan et al., 2009). *Innovative capability* is an imperative element among others, which pertains to leveraging new approaches to achieve organisational goals. Indeed, the ability to create value and creativity are preeminent elements that determine a firm's success (Wang and Ahmed, 2007). Similarly, Crossan and Apaydin (2010) maintain that innovative capability serves as a decisive factor that contributes to business results in a volatile environment. Resting on the cyclical hypothesis (Archibugi et al., 2013), Mensch (1979) argued that innovation was counter-cyclical because most firms took refuge over the course of economic downturns by utilising existing rents. In this situation, the firm's capability to rejuvenate and restructure its competitiveness through innovative funding may benefit from rough patches since firms may be compelled to abandon non-profitable methods and goods (Brzozowski et al., 2019; Caballero and Hammour, 1996). *The adaptive capability* was another important component of the dynamic capabilities related to firms' ability to expeditiously reconfigure and coordinate their resources to promptly respond to business changes (Gibson and Birkinshaw, 2004; Sapienza et al., 2006; Zhou and Li, 2010). *The absorptive capability* was defined as recognising, developing, and capitalising on external knowledge to create valuable new knowledge (Lane et al., 2006; Zhou and Li, 2010). In a dynamic environment, competitive advantage is generated by utilising new knowledge

and converting it into beneficial knowledge (Cadiz et al., 2009; Zhou and Li, 2010). In addition, Lane et al. (2006) suggested that the ability to absorb is also the extent to which firms can employ knowledge acquired from their learning process to advance business strategies.

Other dynamic capabilities have been found in the literature, including *network capability* (Walter et al., 2006) and *market sense capability* (Lindblom et al., 2008; Morgan et al., 2009). *Network capability* refers to firms' ability to initiate and leverage relationships between organisations to obtain diverse resources (Walter et al., 2006). On such a premise, businesses can seize the opportunity to attain a strategic position in the market and establish an inter-organisational network with potential partners, securing and enhancing profits in dynamic business conditions. At the same time, *market sense capability* was described as an organisation's ability to properly grasp its competitors, customers, and business environment (both internal and external) (Lindblom et al., 2008; Morgan et al., 2009). In other words, it was simply the capability to discern external environment vagaries, combined with the utilisation of market information, to accurately predict changes in customer demand (Lindblom et al., 2008). For example, to meet the needs of customers in the current market, which is characterised by digitalisation, firms have been using social networks to attract prospective customers, establish long-term connections with them, and to raise brand recognition (Pandey and Singh, 2012). Thus businesses possessing market sensing competency may precede their rivals in responding to market fluctuations and therefore acquire a leading position ahead of their competition (Morgan et al., 2009).

The dynamic capabilities theory advocated that firms can address any challenge from the external turbulent environment by enhancing their internal capabilities and aligning them with volatile market conditions by employing suitable strategies (Arndt and Pierce, 2018). Consistently, Nandkeolyar et al. (2017) suggested that firms can avoid failure and achieve higher success by managing innovation and product quality in a proactive manner, enhancing financial agility, steering the organisational change, and governing the impacts from the external environment. In this regard, we attempt to clarify the concept of businesses' dynamic capabilities and how they determine strategic plans to maintain companies' functioning abilities in hostile environments, particularly in the context of COVID-19.

2.3 COVID-19 and firm strategy changes

Since the beginning of the COVID-19 outbreak, a significant number of studies have been conducted to investigate how businesses behave in an acute pandemic. For example, by conducting a survey on 5,800 small businesses in the USA, Bartik et al. (2020) asserted that there are three compelling patterns that have been witnessed in how firms respond to the COVID-19 epidemic. The patterns are:

1 a majority of companies are forced to temporarily shut down and large-scale mass-layoffs occur

2 small businesses are financially fragile to the extent that those with meagre financial capital are capable of affording expenses for less than two months, whilst those with more ample cash on hand can endure business until the end of the year

3 enterprises have extensively divergent beliefs regarding the duration of the detrimental effects of COVID-19.

Furthermore, Juergensen et al. (2020) indicated that, in the short term, firms had faced severe disruptions in logistics and supply chain systems as well as a plunge in market demand as a result of immediate lockdown all over the world. Therefore, the COVID-19 pandemic crisis compelled organisations to adjust their business models, leap to digitalisation opportunities, and identify alternatives for methods of product and service delivery to ensure customers' safety (Seetharaman, 2020). Hence, enterprises were required to become nimble and actively develop their advantage competencies, which assisted them in surviving the pressure that the environment exerts on them. Such dynamic capacities pertained to innovating existing products, creating alternative digital products and services, re-designing product delivery channels, determining new strategic positions, identifying new partners in an ecosystem, and transforming existing resources into value-added strategies.

Several academics have developed strategic frameworks that assist firms in addressing crises' challenges (Booth, 2015; Smith, 2005; Wenzel et al., 2020). In particular, in the Wenzel et al. (2020)'s model, four fundamental strategies were revealed to help firms overcome a crisis and remedy its repercussions: *retrenchment*, *persevering*, *innovating* and *exit*. First, the *retrenchment strategy* implied that businesses would adopt initiatives to curb their expenses and complexities (Benner and Zenger, 2016; Pearce and Robbins, 1993). Retrenchment can result in both favourable and unfavourable implications. On the one hand, cost curtailments were particularly beneficial for sustaining financial liquidity and serve as a concrete bedrock for a firm's long-standing continuity (Pearce and Robbins, 1994). Nonetheless, retrenchment measures were contended to be related to diminished performance (Barker and Duhaime, 1997) and possibly accompanied by a change in the company's resources and organisational culture (Ndofor et al., 2013). Second, by employing a *persevering strategy*, firms can continue their businesses or achieve long-term stability despite difficulties and maintain their current position in their product markets. According to Wenzel et al. (2020), the quintessence of this approach was to avoid starting a strategic renovation at inopportune moments, and its viability was associated with the extent to which the crisis endures, such that more prolonged crises result in more limited economic resources. Third, firms that adopt the *innovating strategy* are more likely to be technologically innovative with their products or seek new markets. Extant research has suggested that external forces like cut-throat business competition and technological advancements are known to be the catalysts for new generations of business models (Clauss et al., 2019; Kraus et al., 2020). Especially during a crisis, firms were pushed to become open-minded about innovative ideas (Roy et al., 2018). The innovative strategy might encourage companies in conquering organisational apathy and considering the feasibility of their current strategic management (Ucaktürk et al., 2011). In short, Wenzel et al. (2020) assert that the innovating strategy is a coping technique that has long-term impacts and can revitalise the firm to become sustainably stronger. Finally, *an exit strategy* can be adopted when the other options are condemned to failure. However, an exitting company that fled from an unsuccessful business does liberate fresh resources (Carnahan, 2017). Similarly, exiting can lead to strategy revival and the establishment of a new organisation (Ren et al., 2019).

Table 1 Notable literature on COVID-19 related business problems and firms' coping strategies

	Study	*Subjects*	*Data collection*	*COVID-19 related problem*	*Dynamic capabilities*	*Firms' coping strategies*
1	Nayal et al. (2021)	Senior executives	Semi-structured in-depth interviews	Customer well-being	• Market orientation • Strategic flexibility • Market innovation	Retaining the flock, financial agility, customer management, prepare for new normal, supply chain agility, digitalisation, corporate social responsibility
2	Ai and Peng (2021)	China's high-end equipment manufacturing listed companies	China Stock Market and Accounting Research (CSMAR) database	Innovation performance	• Coordination and integration capability • Learning capability • Organisational reconstruction capability	Innovation strategies
3	Liu and Yang (2021)	Hoteliers	In-depth interviews	Self-service technology-based adjustment	• Sensing capability • Learning capability • Integrating capability • Coordinating capability	Self-service technology-based strategies
4	Wang et al. (2020)		Systematic reviews	Marketing innovation strategies	• Reconfiguration capability • Leveraging capability	Marketing innovation strategies
5	Do et al. (2021)	Firms' key stakeholders	Semi-structured in-depth interviews	Food supply chain (FSC) resilience	• Sensing capability • Seizing capability	Supply chain agility
6	Dyduch et al. (2021)	Micro, small, and medium-sized Polish firms	CAWI method	Value creation and capture process	No	• Exploiting opportunities • Imitating and Innovating • Diversifying products and services • Developing modern technologies • Orchestrating resources

Table 1 Notable literature on COVID-19 related business problems and firms' coping strategies (continued)

	Study	*Subjects*	*Data collection*	*COVID-19 related problem*	*Dynamic capabilities*	*Firms' coping strategies*
7	Chatterjee and Chaudhuri (2021)	Employees engaged in supply chain management process in India	Survey	Supply chain sustainability	• Technology capability • Innovation capability • Relationship management capability	No
8	Dubey et al. (2021)	Senior-level supply chain managers from the Indian auto components manufacturing industry	Qualitative interviews	Firm's operational and financial performance	• Alliance Management Capability (AMC) • AI-driven supply chain analytics capability (AI-SCAC)	No
9	Kraus et al. (2020)	Family firms in five European countries	Semi-structured interviews	Family firm's crisis management	No	• Retrenchment • Preserving • Innovation
10	Breier et al. (2021)	Hospitality firms in Austria	Multiple case study	Hospitality performance	No	Business model innovation (BMI)
11	Sharma et al. (2020)	NASDAQ 100 firms	Twitter data	• Demand-supply mismatch • Technological challenges • Development of a resilient supply chain	No	Supply chain strategies
12	Belhadi et al. (2021)	Participants from the automobile and airline industries	• Phase 1: secondary sources • Phase 2: Survey • Phase 3: Semi-structured interviews	Automobile and airline supply chain	No	Short-term and long-term strategies for each industry, including reactive and proactive strategies
13	Hao et al. (2020)	Hotel industry in China	Systematic review	Hotel industry	No	Recommendations

Problems that have confronted firms during the COVID-19 pandemic and how firms respond to them have attracted several studies (see Table 1). Several noteworthy aspects may be gleaned from the literature review. First, most of the current studies focused on area-specific coping strategies, especially the issues related to marketing, supply chain, and technology. Only a few studies included systematic, appropriate, and most popular crisis response strategies in general. Second, a limited number of studies combined the firm's dynamic capabilities as the fundamental foundation to propose effective coping strategies in confronting COVID-19. To fulfil these gaps, this study endeavours to extend the literature on COVID-19 problems and response strategies by uncovering the problems and consequent coping strategies based on the firm's dynamic capabilities.

3 Methodology

3.1 Qualitative approach

Given that this study aims to explore the impact of the COVID-19 pandemic on the business strategies of SMEs in Vietnam, a qualitative approach based on in-depth interviews was adopted. Qualitative research is a type of scientific investigation that emphasises the comprehensive and holistic understanding of the research issues (Dana and Dumez, 2015) that comply with two specific characteristics, including ontology and epistemology (Groenland and Dana, 2020). It is an extensive and complicated set of techniques used to explain phenomena from the participants' perspective; typically, individuals are studied within their habitual surroundings, which adds valuable and vivid information within the human context. The qualitative study sought to investigate actors or agents in action. It is based on their backgrounds, motivations, the metrics of their behaviours and their interaction patterns. In general, qualitative research entails some sort of direct interplay between the observer and the participants being examined. Along with this interplay, qualitative research seeks to directly explore and explain the peculiar inner world of individuals, which is believed to be a distinguishable strength over the quantitative approach (Groenland and Dana, 2020). Qualitative research comprises several benefits, including the opportunity to obtain information directly from the study subject, which helps to not only minimise measurement errors prevalent in survey methods that frequently necessitates premises, but also diminish the possibility that reviewers give erroneous questions (type III error) and/or incorrectly explain the research results (type IV error) (Dana and Dana, 2005).

Extant research has provided consistent evidence that supports the significant influence of working context or cultural environment on entrepreneurs, suggesting that the context of entrepreneurial behaviours should be taken into account when conducting research concerning entrepreneurship (Bjørklund, 2004; Dana and Dana, 2005; Dana, 2002). In this instance, quantitative research conducted with the entrepreneurs as the sole unit of research in laboratories or field studies is insufficient to provide holistic information to understand entrepreneurship. Instead, scholars should adopt alternative approaches that allow for the consideration of critical aspects of the environment. Dana and Dana (2005) suggest that using a qualitative approach is a viable strategy that is workable in studying SMEs and entrepreneurial issues. This study conducted in-depth semi-structured interviews with managers/owners in several Vietnamese SMEs regarding the effects of the COVID-19 pandemic on their business, their responses, and the

outcomes. Their responses were categorised under the dynamic capabilities theory and four types of strategic responses to crises by Wenzel et al. (2020).

3.2 Data collection

The study applied a purposive sampling technique to recruit respondents, who were required to meet several criteria including:

1 being owner/manager of an SME located in Ho Chi Minh City

2 the SME's operation years from at least two years

3 his/her company has survived after the COVID-19 pandemic during the period from March–May 2020.

Table 2 Data descriptions

No.	*Industry*	*Company size**	*Respondent position*	*Working experience*
1	Service (event)	20	CEO	9 years
2	Service (PR and advertising)	100	Chief digital officer	3.5 years
3	Retails	15	Owner	5 years
4	Service (education)	5	Marketing manager	1 year
5	Manufacturing	35	Marketing manager	3 years
6	Service (F&B)	N/A	Store manager	N/A
7	Service (internet and game shop)	4	Manager	2 years
8	Service (online platform providers)	12	Owner/manager	2.5 years
9	Service (event, PR, marketing)	20	CEO	8 years
10	Service (logistics and transportation)	100	Accounting manager	7 years
11	Service (F&B)	15	Store manager	2 years
12	Trading and retailing	32	CEO	9 years
13	Manufacturing and retailing	70	Owners/CEO	15 years
14	Manufacturing (agriculture)	100	Vice director	8 years
15	Service (F&B)	20	Store manager	5 years

Note: *Number of employees; CEO – chief executive officer.

The semi-structured questionnaire was developed to include questions of the effects of the COVID-19 pandemic on their business, their responses, and the outcomes. The core questions were presented as follows:

1 What difficulties did your business face during the COVID-19 pandemic and the social distancing period?

2 How has your business adjusted its strategy, including:
   - 2.1 inputs and costs
   - 2.2 production/operation
   - 2.3 marketing and sales?

3 How is the application of technology in your business?

4 How difficult is it to change business strategy?

5 How long does it take businesses to adjust their business strategy?

6 How do you evaluate the change process of your business compared to your rivals?

7 How are the current revenue and profit compared to the time before the pandemic?

Four researchers have adopted this questionnaire and seek the appropriate respondents. The list of company industries was updated among the researchers to diversify the range of industries. Table 2 presents the respondents' descriptions.

3.3 Analysis

Given the flexibility and appropriation to various data types, the theory-driven thematic analysis was adopted (Nowell et al., 2017) using MAXQDA 2020. To warrant trustworthiness, six phases had been strictly followed. First, to obtain the data familiarisation, researchers read all the transcripts and obtained the main ideas. Next, a pilot coding was carried out with the open coding process on five companies without the literature review. This process helped to initiate ideas and new themes from the data. Then, the research theme was initiated by having another researcher review the dynamic capabilities theory and four types of strategic responses to the crisis by Wenzel et al. (2020) to create a codebook, which included four main categories (i.e., company demographics, market dynamism, dynamic capabilities, and firm strategy). In the next step, the pilot codes were compared and matched with the codebook categories to review themes. After that, to refine the theme, the sub-categories were defined from the suggestion of both theories and the pilot code. Lastly, categories were reported with definitions and examples (Lincoln and Guba, 1985 in Nowell et al., 2017).

To validate the analysis, coder triangulation had been carried out within the comparison between two researchers, whereas data triangulation was also demonstrated by the number of transcript documents supporting categories (Denzin and Lincoln, 2011). Triangulation, "a kind of standard in qualitative research" [Denzin and Lincoln, (2011), p.799], pertains to the use of a variety of approaches for depicting the same phenomena (Denzin, 2017). There are generally four types of triangulation, including data triangulation (e.g., different dataset at different times, places, and participations), investigator triangulation (e.g., different interviewers, different coders), theory triangulation (i.e., different theoretical viewpoints), and methodological triangulation (Denzin, 2017; Denzin and Lincoln, 2011). The converged triangulation findings can serve to validate the phenomena and reach conclusions. Meanwhile, the divergence in triangulation data can give further explanation and implication possibilities (Denzin and Lincoln, 2011). In this study, data triangulation (i.e., data was collected from different participants at different times and places) and researcher triangulation were applied.

4 Findings

The thematic analysis revealed three main categories (market dynamism, dynamic capabilities, and firm strategy) and 13 sub-categories (see Table 3).

Market dynamism relates to the changes in the business environment, which included the effects of the COVID-19 pandemic, and changes in the demand and innovation

pressure. Among these businesses' environmental changes, the COVID-19 pandemic and related social distancing policies were likely to have substantial negative impacts on all surveyed firms, such as loss of profit, decreased sales, and being forced to shut down temporarily. Besides the effects of the pandemic, the changes in demand also negatively affected the SMEs' behaviour. During the COVID-19 pandemic, the changes in demand originated from the changes in customers' preferences (e.g., using 'packaged coffee, as well as tools for making coffee' instead of hanging out), customers' needs (e.g., using 'regular rice over organic rice'), and decreases in demands for specific services such as studying abroad. Innovation pressure also became one of the main factors influencing SMEs' strategy adaptation. Specifically, the majority of SMEs have been aware of the popularity of social network sites and digital business environments.

Dynamic capabilities included market sense, network capability, firm agility, innovative capability, absorptive capability, and adaptive capability. Market sense refers to the owners/managers' capability to recognise, understand, and even predict changes in their business environment. Most of the survey SMEs performed their market sense capability when they have perceived the changes in the business environment and their competitors' behaviour. Several owners/managers presented their forecast on the pandemic and even posted the statement that "the company can see an opportunity". Besides market sense, the majority of SMEs showed their network capability and firm agility. The literature has indicated that network capability and firm agility were required for firms to overcome the effects of the pandemic (Do et al., 2021; Nayal et al., 2021). Our findings confirmed these studies' results. Additionally, the findings emphasised that SMEs in Vietnam have the agility capability in business goal adjustment and digitalised several parts of their business operation (e.g., marketing, sales). SMEs also perform their network capability when negotiating with their partners (i.e., rental providers, shipping services, payment services and other service providers) to reduce their costs. For the innovative capability, the mix of capabilities has been revealed, including new organisational structure, new customers, new markets, new methods, and new products. Among these, new methods (i.e., new ways to approach and serve customers) and new organisational structure (e.g., arranging work from home, rearranging the workforce and finance) were performed by the majority of the SMEs; whereas, and other innovative capability dimensions were performed in several firms. Finally, SMEs also performed their absorptive and adaptive capabilities. Several firms performed their adaptive capabilities via the capability of combining their resources to identify and deliver their services to customers during the pandemic. For example, several SMEs have moved their business to online stores (e.g., website, retailing platforms) and initiated new products and services (e.g., online consulting, new products appropriate to online stores). Meanwhile, SMEs with absorptive capabilities have learned to think outside knowledge (e.g., from their competitors and/or customers) when operating their business during the COVID-19 pandemic. Specifically, these firms organised employee training, applied online marketing tools, and prepared a risk management plan to deal with the new business environment.

SMEs in Vietnam have adjusted their business strategies to adapt to the changes in the business environment. Four strategies from the literature, including preserving, retrenchment, analysing and innovating, have been carried out but in different manners in practice. During the pandemic, retrenchment or defensive strategy has been the most dominant strategy that most SMEs have followed. Specifically, they attempted to reduce their operating costs by cutting down the number of employees and reducing rental and

administration costs. Meanwhile, SMEs following a retrenchment strategy also maintained their current business by focusing on the current customers. The persevering strategy was the second most popular choice that SMEs used during the pandemic when they just carried out temporary actions such as shutting down their stores for a short time and laying off part-time employees. A worthy note is that this strategy represented the passive reactions of the SMEs with less or no preparation and plan for crisis management. On the opposite side, nearly half of the surveyed SMEs followed an active strategy to deal with the changes in the business environment, namely the innovating strategy and the analysing strategy. For the innovating SMEs, they had taken opportunities to expand their business, such as opening new stores when the rental prices were lowered, creating new alliances, and exploiting the advantages of online channels to approach their customers. Whereas the analysing SMEs have re-analysed their situation and the changes in the business environment before carrying out appropriate actions. However, only two among 15 SMEs carried out this strategy.

Table 3 Main categories, sub-categories, and supporting quotes

Categories	*Supporting quotations*
1 Market dynamism	
1.1 COVID-19 (15)	• "The social distancing has forced retail stores to close at the request of the government." • "The shop has to close down and has absolutely no revenue but has to pay expenses" • "The COVID-19 pandemic has had a huge impact on the store, not only ours but all F&B industry." • "During the time of social distancing, the company temporarily stopped operating"
1.2 Demand (11)	• "After the pandemic, the number of customers has returned, but not as much as before, students who almost returned to their hometown, study online and are no longer at the city " • "Customers changed their buying demand. They started buying packaged coffee, as well as tools for making coffee." • "From the middle of the year to now, thanks to the significant development of the chain of thegioididong, and FPT, the company's revenue has increased dramatically." [*thegioididong, FPT: two giant retailers in electronics and high-tech devices in Vietnam]

Note: The number next to the categories indicates the number of interviews supporting category)

Table 3 Main categories, sub-categories, and supporting quotes (continued)

Categories	*Supporting quotations*
	• "Although rice is a staple, customers still prefer to use regular rice over organic rice. Purchasing power decreased, revenue decreased, and cash flow targets were not met, accordingly"
	• "Due to the complicated global situation, the demand for studying abroad of customers also decreased during that period."
	• "The number of contracts decreased; revenue plummeted"
1.3 Innovation (7)	• "Yes. The company promotes study abroad campaigns through social network sites like Facebook."
	• "The PR and media environment in recent years has had a strong shift to the digital environment"
	• "Applied technology is emotional and effective."
	• "Educating customers into using those technologies."
2 Dynamic capabilities	
2.1 Market sense (15)	• "Businesses still boldly expand more branches and stores by taking advantage of the bargaining position to wait for shopping demand to bounce back when the epidemic situation has been better controlled."
	• "Since Vietnam controlled the epidemic well and people's psychology was also used to the "new normal", the business prospered again".
	• "According to the EPA company, by the end of this year, the pandemic will be under control. This will make the situation of studying abroad also become stable."
	• "In market volatility, the company can see an opportunity."
	• "As far as I can see, all competitors have changed their way of doing business and are very flexible to be able to maintain the whole system and they have done a very good job."
2.2 Network capability (11)	• "Negotiate with suppliers to reduce rental costs"
	• "The business has actively made the proposal in order to get the deposit back from hotels and convention centres."
	• "Cooperating with major payment platforms and major shipping companies to make customer ordering and receiving activities as simple as possible."

Note: The number next to the categories indicates the number of interviews supporting category)

Table 3 Main categories, sub-categories, and supporting quotes (continued)

Categories	*Supporting quotations*
2.3 Firm agility (14)	• "With the advantage of flexible small scale and having been doing business in parallel with traditional and online for a long time, it only takes about 1-2 days for businesses to adjust to the new direction" • "Actually, we're extremely flexible in our business. For example, if we're going to catch the market quite sensitively, we'll turn around..." • "Small should be fast and flexible. Fall fast for learning." • "Young employees are quickly compatible with digital, so it doesn't take long to change and get used to operating in the digital environment." • "Businesses themselves change their goals, adapt to the transformation to survive"
2.4 Innovative capability	
2.4.1 New organisational structure (11)	• "The company has cut 50% employees' salaries during the COVID-19 outbreak months." • "The company has increased the capital from shareholders to be able to maintain operations." • "Fully working from home from April." • "During the pandemic: alliancing to organize online events"
2.4.2 New customers (5)	• "Instead of doing in many areas, then just focus on Hair Salon, Nail Salon. Bring in orders right after the pandemic. (Business owners learn from their own experiences and change strategies)"
2.4.3 New markets (8)	• "Businesses still boldly expand more branches and stores" • "Expanding the study abroad segment to the European market, especially to tuition-free countries such as Finland, the Netherlands, and Germany." • "seeking new customers" • "Focusing on a niche market."
2.4.4 New methods (13)	• "Moving to online and delivery when traditional stores can't open." • "Promotional and marketing activities have been carried out on the social network Facebook." • "Shipping to customers door to reduce marketing costs."

Note: The number next to the categories indicates the number of interviews supporting category)

Table 3 Main categories, sub-categories, and supporting quotes (continued)

Categories	*Supporting quotations*
2.4.5 New products (6)	• "The company adopts a product change strategy" • "For event businesses: adopting online technology. Running both offline and online meetings." • "Offering affordable and low-budget event packages for customers"
2.5 Absorptive capability (11)	• "Considering, adjusting the operation process to be more efficient; training and training skills for employees." • "Technology in advertising and marketing is applied more often during the pandemic period. This is effective." • "Don't just react negatively (firing employees, halting operations)" • "Previously, the company did not have a risk management plan."
2.6 Adaptive capability (10)	• "In terms of online business, there are not many difficulties because before, the business has operated in parallel with traditional and online sales channels." • "The company always keeps optimistic and adapts flexibly to the changes." • "Moving to online channels (pre-existing projects, speeding up completion): building websites, providing customer consulting services."
3 Firm Strategy	
3.1 Persevering strategy (8)	• "The store must be closed for 1-2 weeks." • "The company also had to temporarily suspend operations to observe and had to lay off about 30% of employees." • "At the beginning, the adjustments were just temporary. The goal itself is not achieved."
3.2 Retrenchment strategy (14)	• "Try to cut down the human resources expense and admin costs (space, electricity, water, security...) to optimize finance in line with the epidemic situation." • "Reducing unnecessary expenses of the business." • "Keeping existing customers and better serving these customers."

Note: The number next to the categories indicates the number of interviews supporting category)

Table 3 Main categories, sub-categories, and supporting quotes (continued)

Categories	*Supporting quotations*
3.3 Analysing strategy (2)	• "All pre-existing plans are abandoned; be always in the state of improvising, fighting. Observing customers 'need and try to meet customer needs, no matter online and offline customers." • "The company only got lost in April. After that, revenue increased 35%, surpassing target in the same period last year." • "Company is confident that COVID-19 is no longer a big deal."
3.4 Innovating strategy (7)	• "Still boldly expand more stores even during the pandemic" • "Revenue doubled than expected even the pandemic" • "Apply technology that can help to reduce the labour dependency. We also trained our customers in the early stage of the technology adoption." • "Alliance was established and was fortunate to meet the needs of customers" • "Since the deal has been concluded with the exporter, this year's target has been reached, the post-epidemic situation has become more stable."

Note: The number next to the categories indicates the number of interviews supporting category)

5 Discussion

5.1 Theoretical and practical implications

The present study was designed to explore how SMEs cope with the business environment's adversities generated by the current severe COVID-19 pandemic. First and most importantly, this study contributed significantly to the two prominent theories: dynamic capabilities theory and crisis response strategy. The most important finding to emerge from the analysis was the crucial role of dynamic capabilities, which can help SMEs survive and even develop during an acute social crisis. Besides the dynamic capabilities proposed in Wang and Ahmed (2007)'s model, which included innovative capability, adaptive capability, and absorptive capability, this study also highlighted the roles of other capabilities, such as agility capability (Do et al., 2021), market sense capability (Lindblom et al., 2008), and network capability (Parida et al., 2009). Second, this study's findings partly supported Wenzel et al. (2020)'s strategic model for crisis management. In particular, to encounter adversities emanated from the COVID-19 pandemic, most firms employed retrenchment, persevering, and innovating strategies. However, inconsistent with Wenzel et al. (2020)'s, no firm exited the market. Instead, a few firms engaged with the analysing strategy, defined as re-evaluating the firm's status and the changes in the business environment before taking proper decisions. Finally, even though preceding studies have exerted significant effort on examining firms' dynamic

capabilities and crisis management strategies, scarce research addressed the question of how firms with specific dynamic capabilities employ certain strategies, especially in the context of the COVID-19 pandemic.

This study provided practical implications by suggesting that particular business strategies should be implemented depending on the firm's characteristics. Specifically, firms with vigorous innovative capability and agility capability should employ innovative strategies (i.e., new products, new market, new method) that enable them to exploit opportunities from the uncertain market during a crisis. Meanwhile, firms with less or no preparation plan for crisis management should adopt a persevering strategy by which they implement short-term actions. Moreover, analysing firms should consider the analysing strategy so that they can re-analyse the firm's strategies and consider the dynamic factors in the business environment. Lastly, the remaining firms that do not have the above characteristics should employ retrenchment or defensive strategies to shelter themselves from the detrimental effects of the pandemic and survive through the storm.

5.2 Limitations and future research directions

Although this study primarily aimed to explore the changes in SMEs' strategies in response to the COVID-19 pandemic, several limitations should be indicated. First, a small number of samples in this study might affect the generalisability of the research. Future investigations should expand the analysis with a more significant number of interviewed firms to get more representable insights. Second, the study captured firm strategies in the early stage of the COVID-19 pandemic; however, these strategies might change after several waves of social distancing caused by the pandemic. Accordingly, a longitudinal research design to track the strategy changes in the long-term might be a fruitful avenue for consideration. Finally, this study mainly focused on the organisational factors to analyse SMEs' strategy change, but institutional factors and entrepreneurs' factors have not been considered. Therefore, future studies can expand the research scope by integrating these factors.

For other future research directions, the findings of this study suggested that scholars should attach the three new types of dynamic capabilities (i.e., agility, network, and market sense) to Wang and Ahmed's original framework to have a comprehensive understanding of how firms can utilise their dynamic capabilities to confront a dynamic business environment. Another exciting direction is that scholars interested in examining SME crisis management strategies can include analysing strategies to develop an all-inclusive strategic model. Furthermore, future research should theorise and test the relationships between each type of capability and the strategic decision to deal with crises. In doing so, the effect of dynamic capabilities on crisis-related strategies can be tested in a more systematic and thorough manner. Consequently, the study's results can also provide evidence for how each type of dynamic capability directly influences decision making during the COVID-19 pandemic. The calculation of the strength of the effects is more intuitive, and thus the recommendations provided for practitioners will be more readily accepted.

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